

● ARRAY CREATION & BASICS

1. Create an array of 5 student names.
2. Create an array containing mixed data types.
3. What is the difference between `[]` and `new Array()`?
4. Predict the output:

```
let arr = [1, 2, 3];  
console.log(arr.length);
```

5. Can arrays in JavaScript be resized dynamically? Explain.
6. How do you check if a variable is an array?
7. Predict the output:

```
let a = [];  
a[3] = 10;  
console.log(a.length);
```

8. What happens if you access an index that does not exist?
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● ITERATING ARRAYS (for, forEach, map)

9. Print all elements of an array using for loop.
10. Print only even numbers from an array using for loop.
11. Print sum of all array elements using loop.
12. Use `forEach` to print each element with its index.
13. Can `forEach` be stopped using `break`? Why?
14. Convert all numbers of array into their squares using `map`.
15. What is the difference between `map` and `forEach`?
16. Predict the output:

```
let arr = [1,2,3];  
let res = arr.map(x => x * 2);  
console.log(res);
```

17. Predict the output:

```
let arr = [1,2,3];  
let res = arr.forEach(x => x * 2);  
console.log(res);
```

18. When should you prefer map over for loop?

● ADD / REMOVE ELEMENTS

19. Add an element at the end of array.

20. Remove last element from array.

21. Add an element at the beginning of array.

22. Remove first element from array.

23. Predict the output:

```
let arr = [10,20,30];  
arr.push(40);  
arr.pop();  
console.log(arr);
```

24. Predict the output:

```
let arr = [1,2,3];  
arr.unshift(0);  
arr.shift();  
console.log(arr);
```

● MAP, FILTER, REDUCE (IMPORTANT)

25. Use filter to get all numbers greater than 50.

26. Use filter to get only even numbers.

27. Use map to convert prices into discounted prices.

28. Use reduce to find sum of all numbers.

29. Use reduce to find maximum number.

30. Predict the output:

```
let arr = [1,2,3,4];  
let result = arr.reduce((acc, cur) => acc + cur, 0);  
console.log(result);
```

31. What happens if initial value is not passed to reduce?

32. Explain a real-world use case of reduce.

🔍 SEARCHING IN ARRAYS

33. Use find to get first number greater than 10.

34. Use findIndex to get index of first negative number.

35. What is the difference between find and filter?

36. Predict the output:

```
let arr = [5, 12, 8, 130];  
console.log(arr.find(x => x > 10));
```

37. Check if array contains value 100.

38. Find index of value 25 in array.

39. What is returned if value is not found using indexOf?

🔍 SOME, EVERY, INCLUDES

40. Use some to check if any number is even.

41. Use every to check if all numbers are positive.

42. Predict the output:

```
let arr = [2,4,6];  
console.log(arr.every(x => x % 2 === 0));
```

43. Difference between some and every.

44. Difference between includes and indexOf.

🔍 SLICE vs SPLICE (TRICKY)

45. Extract first 3 elements using slice.

46. Remove 2 elements starting from index 1 using splice.

47. Predict the output:

Java Script Arrays

```
let arr = [1,2,3,4];  
let newArr = arr.slice(1,3);  
console.log(arr, newArr);
```

48. Predict the output:

```
let arr = [1,2,3,4];  
arr.splice(1,2);  
console.log(arr);
```

49. What is the key difference between slice and splice?

● ARRAY DESTRUCTURING

50. Destructure first two elements from an array.

☒ INSTRUCTIONS

- Solve all 50 questions on paper or editor
- Focus on logic, not syntax
- After completion, say:

👉 **“Give me answers for Array questions”**

I will then provide: - Step-by-step solutions - Visual explanations - Common mistakes - Interview tips